SatQuery-Edge

Offline Satellite Intelligence Investigation Report

SIH 26167 · ISRO · Prototype · Offline Mode

Investigation ID	SQE-20260902-GPNE5P
Timestamp	2026-09-02 18:16:12
Intent	bi_temporal_change
Tools Used	change, grounding, geospatial

Query

Identify newly flooded built-up areas and show the highest-confidence regions.

Final Answer

Candidate newly flooded built-up regions detected.

• 2 significant changed region(s) identified.
• Approximately 16.4% of scene shows change.
• Evidence Score: 80/100 — HIGH CONFIDENCE.

Detected change signatures are consistent with new water-like inundation over areas previously identified as built-up terrain. Highest-confidence regions are highlighted in the overlay and mapped.

NOTE: Results are demonstration-grade. Not a scientifically validated remote-sensing product.

Evidence Score

Component	Code	Value	Weight
Model Agreement	M	0.884	0.30
Sensor Reliability	S	0.900	0.20
Spatial Consistency	P	0.733	0.20
Temporal Consistency	T	0.940	0.15
Input Quality	I	0.478	0.15
FINAL SCORE		**80.5 / 100**	

Validation

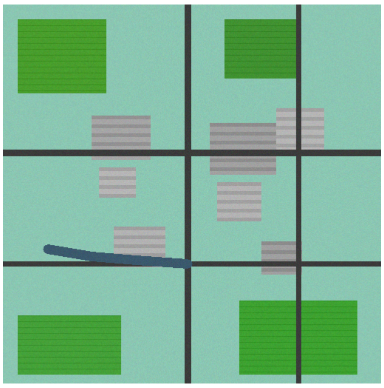
Status: **PASS** ✓
• 2 image(s) detected — count OK.
• Image 1: Quality OK (mean=159, std=35.5).
• Image 2: Quality OK (mean=153, std=36.3).

• Image pair: identical dimensions — perfect alignment.

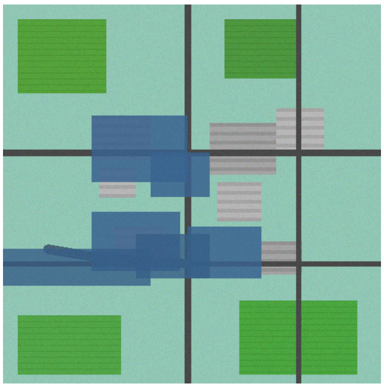
Detected Regions & Approximate Coordinates

#	Label	Confidence	Area (km²)	Demo Geore
1	candidate changed region	0.95	18.215	Yes
2	candidate changed region	0.83	9.490	Yes
3	candidate flooded built-up are	0.97	128.691	Yes

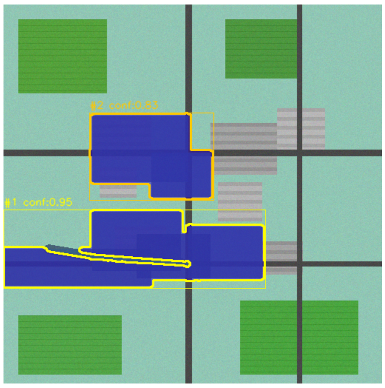
Key Visual Evidence



Pre-Event Image



Post-Event Image



Change Overlay

Specialist Tool Outputs

CHANGE — SUCCESS — 2 region(s)

```
Detected 2 significant candidate changed region(s).

Largest region:
Approximate change area: 9.9% of scene.
Spectral Change Confidence: 0.95
Center: (0.37, 0.66) (normalized pixel space)

Total changed pixels: ~16.3% of observation area.

Remote Sensing Analysis (NDWI / Inundation Dynamics):
Candidate newly inundated / flooded regions where spectral reflectivity
changed significantly be
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GROUNDING — SUCCESS — 1 region(s)

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Grounding DINO + SAM Engine found 1 candidate region(s) matching target 'flooded built-up
areas'.
Highest confidence: 0.97.
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GEOSPATIAL — SUCCESS — 0 region(s)

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Geospatial conversion complete.
Converted 3 region(s) to geographic coordinates.
Total approximate area: 156.40 km².

■ Approximate demo georeferencing applied.
Demo center: (20.296°N, 85.824°E)
Coordinates are NOT derived from real GeoTIFF metadata.
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Execution Trace

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[18:16:12] QUERY RECEIVED: Identify newly flooded built-up areas and show the
highest-confidence regions.
[18:16:12] ROUTER DECISION: Intent: bi_temporal_change
Selected tools: change, grounding, geospatial
Images required: 2
[18:16:12] VALIDATION: Status: PASS
2 image(s) detected — count OK.
Image 1: Quality OK (mean=159, std=35.5).
Image 2: Quality OK (mean=153, std=36.3).
[18:16:12] TOOL EXECUTING: change → started
[18:16:12] TOOL OUTPUT (change): Success — 2 region(s) detected
Detected 2 significant candidate changed region(s).

Largest region:
[18:16:12] TOOL EXECUTING: grounding → started
[18:16:12] TOOL OUTPUT (grounding): Success — 1 region(s) detected
Grounding DINO + SAM Engine found 1 candidate region(s) matching target 'flooded built-up
areas'.
Highest confidence: 0.97.
[18:16:12] TOOL EXECUTING: geospatial → started
[18:16:12] TOOL OUTPUT (geospatial): Success — 0 region(s) detected
Geospatial conversion complete.
Converted 3 region(s) to geographic coordinates.
Total approximate area: 156.40 km².
[18:16:12] EVIDENCE FUSION: Score: 80.5/100
M=0.884
S=0.900
P=0.733
[18:16:12] FINAL SCORE: 80/100
[18:16:12] ANSWER: Candidate newly flooded built-up regions detected.

• 2 significant changed region(s) identified.
• Approximately 16.4%
```

Tool Versions

Component	Version
SatQuery-Edge Prototype	1.0.0-SIH26167
Python OpenCV	4.11.0
NumPy	1.26.4
Streamlit	1.62.0
Report Engine	ReportLab PDF
Mode	OFFLINE — No external APIs

NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. — SIH 26167 · ISRO · SatQuery-Edge